Worksheet 2-3 B Tangent Lines #1
AP Calculus Name_____

1. A function $f(x)$ is continuous and differentiable for all values of x. It is known that $f(-2) = 5$ and $f'(-2) = -3$. Write an equation of the line tangent to $f(x)$ at $x = -2$.

2. The function g is defined as $g(x) = 2x^3 + x - 2$. It is known that the derivative of g is defined as $g'(x) = 6x^2 + 1$. Write an equation of the line tangent to $g(x)$ at $x = 2$.

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x	-4	-1	2	5
$h(x)$	1	-2	0	7
$h'(x)$	-7	3	2	0

The function $h(x)$ is differentiable for all values of x with selected values of h and h' in the table above.

3. Write the equation of the line tangent to $h(x)$ at $x = 2$.

4. Write the equation of the line tangent to $h(x)$ at $x = -1$.

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5. (Calculator Active): The function r is defined as $r(x) = \cos(\ln(x^2 + 4))$. Write the equation of the line tangent to $r(x)$ at $x = -3$.

x	-3	1	4	7
$f(x)$	1	4	-2	-3
$f'(x)$	0	2	-1	6
$g(x)$	7	-3	4	-5
$g'(x)$	2	-5	3	7

The functions $f(x)$ and $g(x)$ are differentiable for all values of x with selected values of $f, f', g,$ and g' above.

6. Let $k(x) = f(x)g(x)$. It is known that $k'(x) = f(x)g'(x) + f'(x)g(x)$. Write the equation of the line tangent to $k(x)$ at $x = 1$.

7. Let $p(x) = \frac{f(x)}{g(x)}$. It is known that $p'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$. Write the equation of the line tangent to $p(x)$ at $x = 4$.

8. Let $m(x) = f(g(x))$. It is known that $m'(x) = f'\big(g(x)\big)g'(x)$. Write the equation of the line tangent to $m(x)$ at $x = -3$.

9. Let $q(x) = g\big(f(x)\big)$. It is known that $q'(x) = g'\big(f(x)\big)f'(x)$. Write the equation of the line tangent to $q(x)$ at $x = 7$.